

Variables and Strings

Introduction to computer programming

Management and Artificial Intelligence



Variables = a Name for Values

The equal sign `=` serves for **assigning** a variable name to a value:

```
>>> pi = 3.14
```

where `pi` is the *variable name* and `3.14` is the *value*.

Recall: The equal sign `=` serves for assignments. The equality check reads as `==`.

Retrieve the value associated to a variable by querying the name on the prompt:

```
>>> pi          # input
3.14           # output
```

- All values are stored in the computer memory
- The assignment operation binds a name to a value

Variable names:

- must be meaningful
- can be as long as you like
- can contain both letters and numbers
- cannot start with a number
- can contain underscores (i.e., `_`)
- convention: use only lower letters
- Python-reserved keywords are not allowed:

False	class	finally	is	return
None	continue	for	lambda	try
True	def	from	nonlocal	while
and	del	global	not	with
as	elif	if	or	yield
assert	else	import	pass	break
except	in	raise		

Why variable names?

- reuse **names** instead of **values** throughout the code
- store (temporarily!) values and results
- easier to change the code later

Examples:

```
>>> pi=3.14
>>> radius=2.2
>>> # calculate the area          <-- this is a comment!
>>> area = pi * (radius**2)
>>> # calculate the diameter
>>> diameter = radius*2
```

Syntax of assignments:

- rightmost term of assignment: expression that yields a value
- leftmost term of assignment: variable name

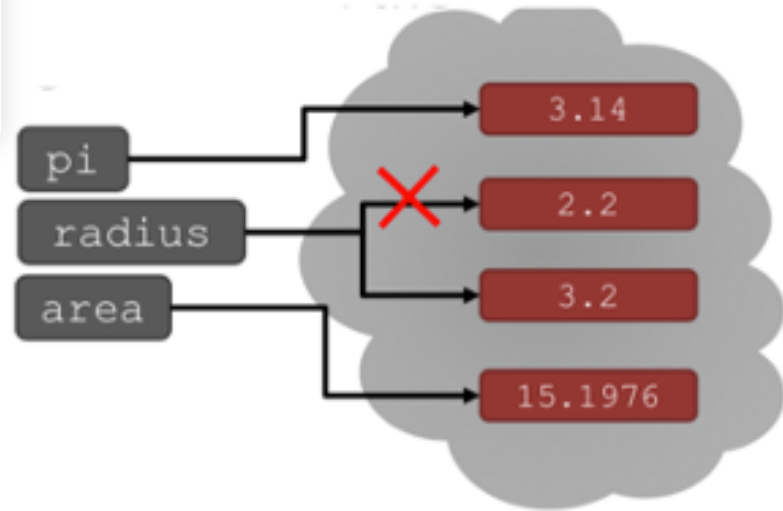
Variables can be **changed** (or **re-bound**) using new assignments on the same variable name.

Warning: The old value might still be in the computer memory, but you lose the handle for it.

```
>>> pi=3.14
>>> radius=2.2
>>> area = pi * (radius**2)
>>> radius = radius+1
```

Hint: `radius = radius+1` can be shortcut to `radius+=1`.

Note: `area` does not change its value.



Dynamic Typing

Python supports **dynamic typing**.

- variables can hold values of any type
- variables can hold different types throughout the code

The following is perfectly valid in Python:

```
>>> x=3           # x now contains an int
>>> ...
>>> x=9.53642     # x now contains a float
```

Recall: You can always use `type(x)` to find the type of variable ``x``.

You can use `type()` for comparison and conversion

```
>>> x=3
>>> type(x)==int
True
>>> x=float(x)
>>> type(x)==float
True
```

Strings: a Warmup

Strings are objects that can host letters, numbers, spaces, special characters. Strings can be created by enclosing stuff in quotation marks or single quotes:

```
>>> hi='hello there' # or "hello there"
```

You can **concatenate** strings using the `+` operator:

```
>>> name='alessio'
>>> greeting=hi + name
>>> greeting
'hello therealessio'
>>> greeting=hi + " " + name
>>> greeting
'hello there alessio'
```

You can **replicate** strings using the `*` operator:

```
>>> name='alessio'
>>> silly=name*3
>>> silly
'alessioalessioalessio'
```

Single quotes or double quotes can be (in principle) used interchangeably. For example:

```
>>> s1 = 'hello students'  
>>> s2 = "hello students"
```

yield two identical strings.

Problem: What if my string contains itself a single quote (e.g., *Let's go*)?

If we do something like

```
>>> s = 'Let's go'  
SyntaxError: invalid syntax
```

we have an unterminated string syntax error.

Solution: Wrap your string with single quotes between double quotes!

```
>>> s = "Let's go"
```

(Opposite) problem: What if my string contains itself a double quote (e.g., *I said "Hello"*)?

If we do something like

```
>>> s = "I said "Hello""  
SyntaxError: invalid syntax
```

we have another syntax error, equivalent to the previous one.

Solution: Wrap your string with double quotes between single quotes!

```
>>> s = 'I said "Hello"'
```

(Yet another) problem: What if my string contains both single quotes and double quotes (e.g., *I said "What's going on"*)?

We cannot do any of the following

```
>>> s = "I said "What's going on""
SyntaxError: invalid syntax
>>> s = 'I said "What's going on"'
SyntaxError: invalid syntax
```

because of syntax errors, equivalent to the previous ones.

Solution: Wrap your string between triple single quotes!

```
>>> s = '''I said "What's going on"'''
```

Escape Sequences

We can solve the above problem also thanks to **escape sequences**.

Escape sequences are brief sequences of characters that start with a backslash and are interpreted differently.

There exist escape sequences to display single quotes (`\'`) and double quotes (`\"`) ... and to display the backslash itself (`\\`):

- if we 'shield' single or double quotes with a backslash, they are interpreted as characters rather than string delimiters;
- if we 'shield' the backslash with a backslash, it is interpreted as character rather than 'this the beginning of an escape sequence'.

Let's try the following:

```
>>> s1 = 'I said "What\'s going on"'
>>> s2 = "I said \"What's going on\""
>>> s3 = "Hello\\Students"
```

There exist escape sequences also for string formatting.

Let's see the most important ones:

- new line `\n`, which changes the line and takes the cursor to the beginning of a new line
- tab space `\t`, which leaves a horizontal tab space
- carriage return `\r`, which moves the cursor back to the beginning of the line
- alarm `\a`, which plays the system alarm (bell)
- form feed `\v` or `\f`, which breaks the line

Let's see them in action:

```
>>> s1 = 'Hello\nStudents'
>>> print(s1)
Hello
Students
>>> s2 = 'Hello\tStudents'
>>> print(s2)
Hello    Students
>>> s3 = 'Hello\rStudents'
>>> print(s3)
Students
>>> print('\a')
```

Input/Output

- Show some stuff to the command line
- Universal (and very flexible) keyword: `print()`

Examples:

```
>>> print('apple', 'banana') # two input arguments
apple banana
>>> print('apple' + 'banana') # one input argument
applebanana
```

By default, multiple input arguments are separated by space () and the line terminates with a newline (`\n`). You can change that via `sep` and `end` .

Example:

```
>>> a=5
>>> print("a=", a, 10, "bye", sep='|', end='\n\n\n')
a=|5|10|bye

>>>
```

Hence, there are two strategies to print something on screen:

1. pass multiple arguments to `print()` and it automatically takes care of printing strings, numbers and different data types
2. build your own string and just `print()` it.

The second case deserves further observations.

What happens if we run the following code?

```
>>> age=5
>>> name='Eric'
>>> print("My name is "+name+" and I'm "+str(age)+" years old.")
My name is Eric and I'm 5 years old.
```

It will error because we cannot concatenate (i.e., `+`) strings and numbers together. Hence, we must convert `age` to string (i.e., using `str()`).

String Formatting

Luckily, Python has several string formatting shortcuts that allow us to create strings

- without manual concatenation and casting
- in a very elegant and clean way.

Let's see the old-school way (C-like).

What happens if we run the following code?

```
>>> n1 = 5
>>> n2 = 1234.56789
>>> print("The value of n1 is %d" % n1)
The value of n1 is 5
>>> print("The value of n2 is %d" % n2)
The value of n2 is 1234
>>> print("The value of n1 is %f" % n1)
The value of n1 is 5.000000
>>> print("The value of n2 is %f" % n2)
The value of n2 is 1234.567890
```

In the string we are building, we can see that:

- the percentage sign marks the place where the value has to be inserted
- `%` is followed by the conversion format, i.e.:
 - `%d` stands for decimal (i.e., integer)
 - `%f` stands for floating point
- the format automatically performs casting
 - `n1` is an integer, but it renders as float if we use `%f`
 - `n2` is a float, but it renders as integer if we use `%d`

Outside the string we are building, we note that:

- `%` is followed by the variable name (or value) we want to plug inside the string

Note: We can personalize `%f` by specifying the number of digits before and after the decimal separator. That is, `%.xf` yields a floating point number with *x* digits before and *y* digits after the comma. *x* can be omitted.

We can build strings with an arbitrary number of placeholders.

Let's run the following code

```
>>> name = "Eric"
>>> age = 5
>>> print("My name is %s and I'm %d years old." %(name, age))
My name is Eric and I'm 5 years old.
```

We note that:

- we have two placeholders inside the string
 - `%s` stands for strings
 - `%d` follows from previous slide
- outside the string, we have our variables/values collected in round brackets (That's a *tuple*, we will see tuples later in the course.)
- the order of the placeholders must match the ones in the tuple

Note: `%s` can be used for strings and for all objects that support a string conversion, e.g., numbers.

Input: `input()`

- Ask the user for some input via keyboard
- Keyword: `input('...')`

How does it work?

1. prints whatever is in the quotes
2. user types something and press Enter
3. binds the value to a variable

Example:

```
>>> text=input('Type something: ')\nType something: hello\n>>> print(text*3)\nhellohellohello
```

Warning: `input()` always returns a string. If you have to play with numbers, you must cast.

Example:

```
>>> num=input('Enter a number: ')
Enter a number: 5
>>> num
'5'
>>> type(num)
<class 'str'>
>>> num = int(num)
>>> type(num)
<class 'int'>
```

Some Recaps

- **Type: `int`**
 - values: integers
 - operations: `+`, `-`, `/`, `//`, `*`, `%`, `**`, ...
- **Type: `float`**
 - values: real numbers
 - operations: `+`, `-`, `/`, `//`, `*`, `%`, `**`, ...
- **Type: `bool`**
 - values: `True` and `False`
 - operations: `or`, `and`, `not`
- **Type: `str`**
 - values: string literals
 - operations: `+`, `*`, ...

Note: We will see more (non-scalar) data types later in the course.

Interactive vs. Script

What happens if you type this code in the interpreter or write a script and execute it?

```
>>> miles = 11.41
>>> miles
11.41
>>> miles * 2
22.82
```

Interactive	Script
Assignments are silent	Assignments are silent
Variable query returns its content	Variable query is silent
Expressions are returned in real-time	Expressions are silent

Note: If you want your script to show something, you must use ``print()``.

Expressions vs. Statements

- **Expressions:** **Represent** something
 - Python *evaluates* expressions
 - end-result is a value

Examples:

```
>>> 2.3          # a plain literal
2.3
>>> (7*3+1)-4    # 4 literals and some operators
18
```

- **Statements:** **Do** something
 - Python *executes* statements
 - no need to return a value

Examples:

```
>>> print("Hi there!")
Hi there!
>>> import sys
```

